4 (a) (i) State **one** difference and **one** similarity between the oscillations of a stationary sound wave and a progressive sound wave.

Difference: .....

.....

.....

Similarity: .....

.....

.....

[2]

(ii) Fig. 4.1 represents the displacement-time graph of two progressive sound waves travelling together through the same space.

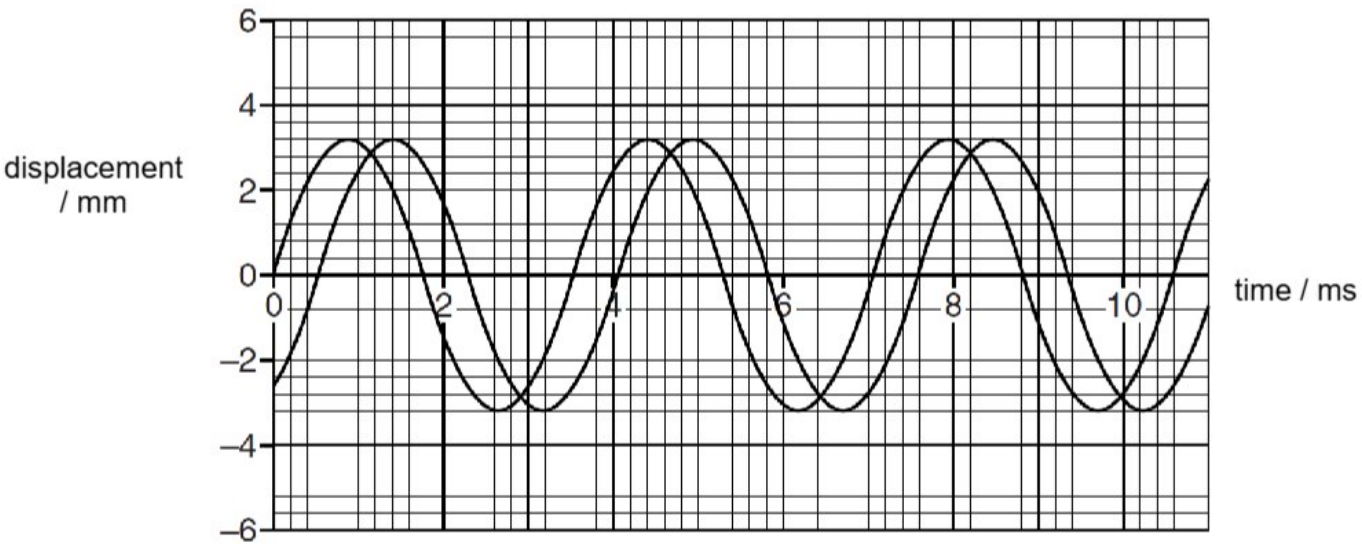


Fig. 4.1

1. State and explain whether Fig. 4.1 shows that the waves are coherent.

.....

.....

.....[1]

2. Use Fig. 4.1 to determine the phase difference between the waves.

phase difference = ... rad [1]

(iii) A sound wave travels along the axis of a tube that is open at both ends, as shown in Fig. 4.2. A stationary sound wave is formed in the tube.

The lowest frequency possible for the stationary sound wave in this tube is *f*.

On Fig. 4.2, mark all the positions of the displacement nodes (use the letter N) and the displacement antinodes (use the letter A) of a stationary sound wave for a frequency of 2*f*.

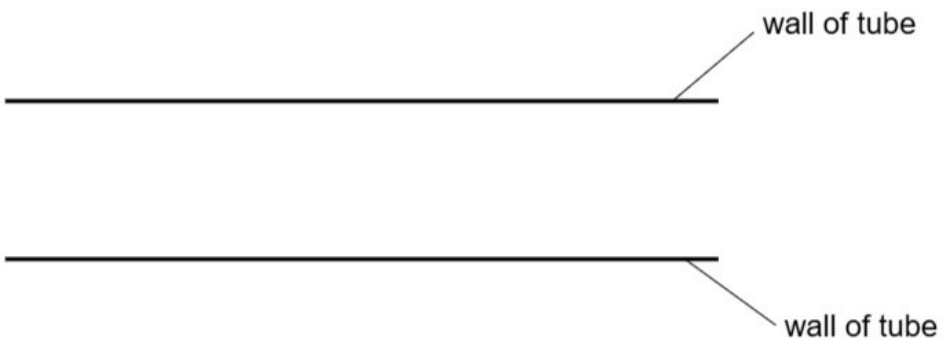


Fig. 4.2

[1]

(iv) With reference to the movement of air molecules on either side of a displacement node, explain why there is maximum variation in pressure at the nodes.

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.....[2]

(b) Fig. 4.3 shows the arrangement for an experiment involving superposition of microwaves.

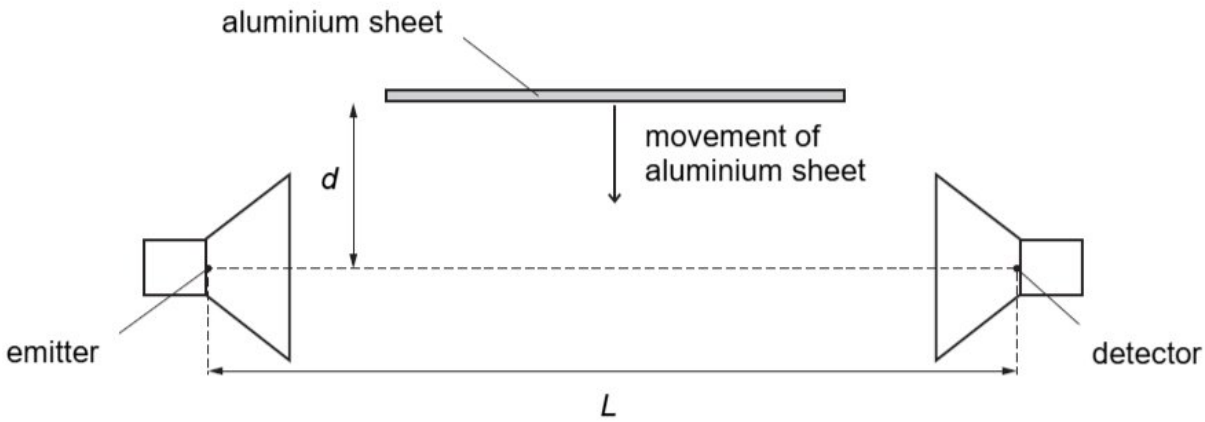


Fig. 4.3 (not to scale)

A microwave emitter and detector face each other with a distance *L* apart. A reflecting aluminium sheet is placed at a vertical distance *d* above the direct path between the emitter and the detector.

(i) When the aluminium sheet is moved in the direction shown by the arrow, the intensity of microwaves received by the detector fluctuates.

Describe and explain this variation in the microwave intensity.

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.....[2]

(ii) As d decreases, the microwave detector has a maximum reading at d = 21.2 cm and the next maximum at d = 17.7 cm. The distance L stays constant at 80.0 cm.

Determine the wavelength λ of the microwaves.

λ = ... m [3]

(iii) Microwaves undergo a phase change of π rad when they reflect.

When d is much less than the wavelength of the microwaves, it is observed that the detector reading is a minimum.

Suggest why this observation demonstrates that microwaves must undergo a phase change on reflection from the aluminium sheet.

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.....[2]

[Total: 14]